6 **(a)** State Faraday's law of electromagnetic induction.

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.....

..... [1]

(b) A solenoid S has a small coil C placed near to one of its ends, as shown in Fig. 6.1.

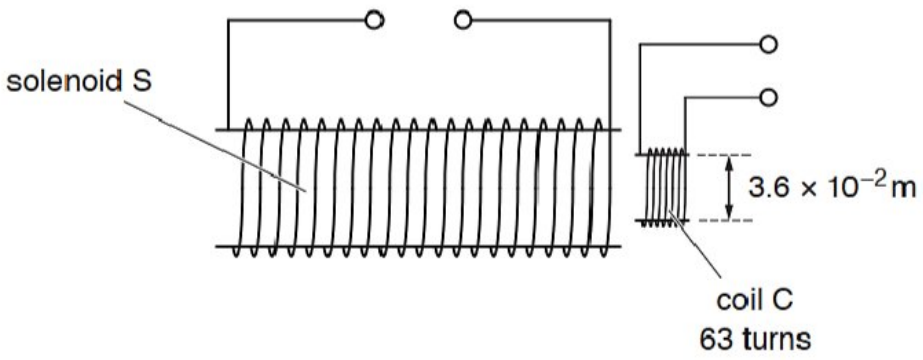


Fig. 6.1

The solenoid S produces a uniform magnetic field of flux density B, in tesla, in the region of coil C given by the expression

$$B = 9.4 \times 10^{-4} I$$

where I is the current, in ampere, in the solenoid S.

The variation with time t of the current I in solenoid S is shown in Fig. 6.2.

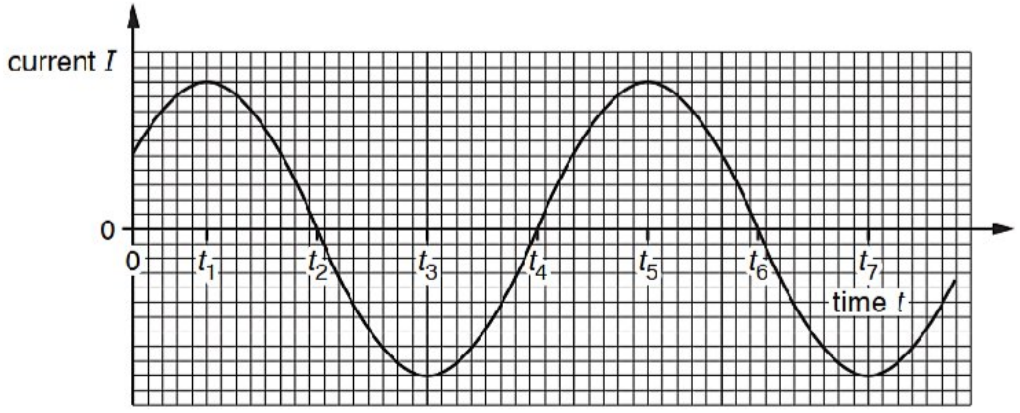


Fig. 6.2

(i) With reference to Fig. 6.2, state two times at which there is no electromotive force (e.m.f.) induced in coil C.

Explain your answer with reference to Fig. 6.2.

time …….. and ……

……

…… [2]

(ii) At a time between t_1 and t_2, state whether the magnetic field produced by the induced current in coil C is in the same direction as or opposite to the magnetic field produced by the solenoid. Explain your answer.

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..... [2]

(c) The alternating current in the solenoid S in **(b)** is replaced by a constant current of 5.0 A.

This current is then reversed in a time of 6.0 ms, while maintaining the same magnitude.

Calculate the average e.m.f. induced in coil C during this period.

e.m.f. induced = ... V [3]